The Fundamental Theorem of Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: evaluating definite integrals

Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$ $[x^2+x]_1^3 = 12-2 = 10.$

b $\int_0^{\pi} \sin x\,dx$ $[-\cos x]_0^{\pi} = 1-(-1) = 2.$

c $\int_1^4 \sqrt{x}\,dx$ $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3}-\frac{2}{3} = \frac{14}{3}.$

(a) $[x^2+x]_1^3 = 12-2 = 10.$; (b) $[-\cos x]_0^{\pi} = 1-(-1) = 2.$; (c) $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3}-\frac{2}{3} = \frac{14}{3}.$

2 [Foundational] • Target: evaluating definite integrals

Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.

$\left[\frac{1}{2}e^{2x}\right]_0^1 = \frac{1}{2}(e^2-1) \approx 3.195.$

3 [Foundational] • Target: differentiating an integral

Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.

$= x^3+1$ (directly, by FTC Part 1).

4 [Foundational] • Target: differentiating an integral

Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.

$= \sin(x^2)\cdot 2x = 2x\sin(x^2).$

5 [Foundational] • Target: Average value and applications

Find the average value of $f(x) = x^2$ on $[0, 3]$.

$\frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3}\cdot 9 = 3.$

6 [Foundational] • Target: Average value and applications

A car's velocity is $v(t) = 3t^2+2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.

Distance $= \int_0^4 (3t^2+2)\,dt = [t^3+2t]_0^4 = 64+8 = 72$ m. Average velocity $= \frac{72}{4} = 18$ m/s.

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7 [Foundational] • Target: Further FTC practice

Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.

$\int_x^5 \cos(t^2)\,dt = -\int_5^x \cos(t^2)\,dt$, so the derivative is $-\cos(x^2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Further FTC practice

Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.

$\frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{2}{\pi} \approx 0.637$.

9 [Intermediate] • Target: Mean Value Theorem for integrals

Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

Average value $= \frac{1}{3}\int_0^3 x^2\,dx = 3$ (found earlier). Solve $c^2 = 3$: $c = \sqrt{3}$ (taking the positive root within $[0, 3]$).

10 [Intermediate] • Target: Mean Value Theorem for integrals

Evaluate $\int_2^5 f(x)\,dx$ given that the average value of f on $[2, 5]$ is 7.

Average value $= \frac{1}{5-2}\int_2^5 f(x)\,dx = 7$, so $\int_2^5 f(x)\,dx = 7(3) = 21$.

11 [Intermediate] • Target: Integrals of piecewise functions

Evaluate $\int_0^4 f(x)\,dx$ where $f(x) = x$ for $x \le 2$ and $f(x) = 2x - 2$ for $x > 2$.

$\int_0^2 x\,dx + \int_2^4 (2x-2)\,dx = \left[\frac{x^2}{2}\right]_0^2 + [x^2 - 2x]_2^4 = 2 + (8 - 0) = 10$.

12 [Intermediate] • Target: Visualizing the two parts of FTC together

For $g(x) = \int_0^x (t^2 - 2)\,dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.

$g'(x) = x^2 - 2$ (FTC Part 1). Setting $g'(x) = 0$: $x = \pm\sqrt{2}$. Since g' changes from negative to positive at $x = \sqrt{2}$ (for $x > 0$), g has a local minimum at $x = \sqrt{2}$.

13 [Intermediate] • Target: A capstone FTC application

The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.

Volume $= \int_0^\pi (5 + 2\sin t)dt = [5t - 2\cos t]_0^\pi = (5\pi + 2) - (0 - 2) = 5\pi + 4 \approx 19.71$ liters.

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14 [Intermediate] • Target: FTC Part 1 with more complex upper limits

Find $\frac{d}{dx}\int_{x^2}^{5}\cos(t)\,dt$, using $\int_{x^2}^{5} = -\int_{5}^{x^2}$ before differentiating.

$\int_{x^2}^{5}\cos t\,dt = -\int_{5}^{x^2}\cos t\,dt$, so the derivative is $-\cos(x^2)\cdot 2x = -2x\cos(x^2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Net change and the FTC • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.

Net change $= -\int_0^5 (3t+2)\,dt = -\left[\frac{3t^2}{2} + 2t\right]_0^5 = -(37.5 + 10) = -47.5$ liters. Amount remaining $= 200 - 47.5 = 152.5$ liters.

16 [Challenging] • Target: Visualizing FTC Part 2 as area • Cross-Lesson: 15.01 Double Integrals

The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1)\,dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).

$\left[\frac{x^3}{3} - x\right]_0^3 = (9 - 3) - 0 = 6$. This is the SIGNED area: the small region below the axis on $[0, 1]$ is subtracted from the larger region above the axis on $[1, 3]$, so 6 is a net value, not the total unsigned area.

17 [Challenging] • Target: Combining FTC Part 1 and Part 2 in one problem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Let $F(x) = \int_1^x (3t^2 - 2)\,dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.

$F(2) = [t^3 - 2t]_1^2 = (8 - 4) - (1 - 2) = 4 - (-1) = 5$. By Part 1, $F'(x) = 3x^2 - 2$, so $F'(2) = 12 - 2 = 10$, which matches the integrand $3t^2 - 2$ evaluated at $t = 2$: $3(4) - 2 = 10$ ✓.

18 [Challenging] • Target: Integrals of piecewise functions • Cross-Lesson: 2.01 Evaluating Limits

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Visualizing the two parts of FTC together • Cross-Lesson: 2.02 Continuity

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$

Let $u = \sin(x^2)$. Differentiating gives $du = \cos(x^2)\cdot 2x\,dx = 2x\cos(x^2)\,dx \Rightarrow x\cos(x^2)\,dx = \frac{1}{2}du$. Transform the integration limits:Lower limit $x = 0 \Rightarrow u = \sin(0) = 0$.Upper limit $x = \sqrt{\pi/2} \Rightarrow u = \sin((\sqrt{\pi/2})^2) = \sin(\pi/2) = 1$.Substitute into the integral.

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20 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 3.01 The Derivative from First Principles

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2 \cos x\,dx = [x^2 \sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x \sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x \sin x\,dx = \frac{\pi^2}{4} - 2J.$ 2. Second IBP on $J = \int_0^{\pi/2} x \sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: evaluating definite integrals

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^{\pi} \cos x\,dx$:

a. 0 ✓

b. 2

c. -2

d. 1

Applying Fundamental Theorem of Calculus principles, $[\sin x]_0^{\pi} = 0 - 0 = 0$. Therefore, the verified correct option is 0.

22 [Foundational] • Target: differentiating an integral

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique – – any antiderivative works, since they differ by a constant ✓

c. bounded on $[a, b]$

d. differentiable everywhere

Any two antiderivatives of f differ by a constant C, which cancels in $F(b) - F(a)$.

23 [Foundational] • Target: Average value and applications

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b - a)$ for some $c \in (a, b)$. Requires f to be:

a. positive

b. integrable and bounded

c. **continuous** ✓

d. analytic

MVT for integrals: f continuous on $[a, b]$.

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24 [Foundational] • Target: Further FTC practice

Proof of FTC: $F(x) = \int_a^x f$. Key step: $F(x+h) - F(x) = \int_x^{x+h} f(t)dt \approx f(x) \cdot h$ by:

a. **MVT for integrals ✓**

b. L'Hôpital

c. product rule

d. chain rule

MVT for integrals gives $\approx f(\xi) \cdot h \to f(x) \cdot h$.

25 [Foundational] • Target: Mean Value Theorem for integrals

FTC Part 1 is proved by constructing:

a. any antiderivative

b. $G(x) = \int_a^x f\,dt$ **and showing** $G' = f$ **✓**

c. Taylor series

d. Riemann sums

Key step: $\lim_{h \to 0} \frac{G(x+h) - G(x)}{h} = f(x)$.

26 [Foundational] • Target: Integrals of piecewise functions

Net Change Theorem states $\int_a^b f'\,dx = f(b) - f(a)$. This is:

a. MVT

b. **FTC Part 2 ✓**

c. EVT

d. Rolle's Theorem

It is FTC Part 2, also called the Net Change Theorem.

27 [Foundational] • Target: Visualizing the two parts of FTC together

FTC Part 1 shows every continuous function has an:

a. **antiderivative ✓**

b. derivative everywhere

c. limit

d. inflection point

Constructs antiderivative $G(x) = \int_a^x f\,dt$.

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28 [Foundational] • Target: A capstone FTC application

A particle has velocity $v(t) = 2t$ m/s. Find its net displacement over $[1, 4]$.

a. 17

b. **15** ✓

c. 13

d. -15

Net displacement $= \int_1^4 2t\,dt = 15$ m.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: FTC Part 1 with more complex upper limits

Find the area of the region bounded by $y = 3x$ and $y = x^2$ in the first quadrant.

a. $\frac{27}{3}$

b. $\frac{27}{6}$ ✓

c. $\frac{27}{2}$

d. $\frac{33}{6}$

Intersections at $x = 0$ and $x = 3$. Area $= \int_0^3 (3x - x^2)\,dx = \left[\frac{3}{2}x^2 - \frac{x^3}{3}\right]_0^3 = \frac{27}{6}$.

30 [Intermediate] • Target: Net change and the FTC

Evaluate the definite integral of the constant function: $\int_8^{11} 21\,dx$.

a. 21

b. 84

c. 42

d. **63** ✓

The antiderivative is $21x$. Evaluating from 8 to 11 gives $21(11 - 8) = 63$.

31 [Intermediate] • Target: Visualizing FTC Part 2 as area

FTC allows computing $\int_0^1 x^{100}\,dx$ via:

a. L'Hôpital

b. **power rule antiderivative** ✓

c. IBP

d. squeeze theorem

Applying Fundamental Theorem principles, Antiderivative $x^{101}/101$. Therefore, the verified correct option is power rule antiderivative.

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32 [Intermediate] • Target: Combining FTC Part 1 and Part 2 in one problem

Average value of f on $[a, b]$:

a. $\frac{f(a)+f(b)}{2}$

b. $\frac{1}{b-a}\int_a^b f\,dx$ ✓

c. $\int_a^b f\,dx$

d. $f\left(\frac{a+b}{2}\right)$

Applying Fundamental Theorem principles, Average value formula. Therefore, the verified correct option is $\frac{1}{b-a}\int_a^b f\,dx$.

33 [Intermediate] • Target: evaluating definite integrals

Area under one arch of $y = \sin x$ (from 0 to π):

a. 0

b. 2 ✓

c. 1

d. π

Applying Applications of Integration principles, $\int_0^{\pi} \sin x\,dx = 2$. Therefore, the verified correct option is 2.

34 [Intermediate] • Target: differentiating an integral

Taylor integral remainder: $R_n = \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t)\,dt$. Requires $f \in$

a. C^0

b. C^n ✓

c. C^∞

d. Lipschitz

Applying Fundamental Theorem principles, n-th derivative must exist. Therefore, the verified correct option is C^n.

35 [Intermediate] • Target: Average value and applications

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4 ✓

d. 8

Applying Fundamental Theorem principles, $2\int_0^2 x\,dx = 2 \cdot 2 = 4$. Therefore, the verified correct option is 4.

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36 [Intermediate] • Target: Further FTC practice

Find the value of c guaranteed by the Mean Value Theorem for $f(x) = \frac{1}{x}$ on $[1, 16]$.

a. $\frac{17}{2}$

b. 8

c. 4 ✓

d. $\frac{19}{2}$

The secant slope is $-\frac{1}{16}$. Solving $-\frac{1}{c^2} = -\frac{1}{16}$ in the interval gives $c = 4$.

37 [Intermediate] • Target: Mean Value Theorem for integrals

$v(t) = 3t^2 - 2t$ on $[0, 2]$. Net displacement:

a. 2

b. 4 ✓

c. 6

d. 8

Applying Fundamental Theorem principles, $[t^3 - t^2]_0^2 = 8 - 4 = 4$. Therefore, the verified correct option is 4.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Integrals of piecewise functions • Cross-Lesson: 2.01 Evaluating Limits

Net change: if velocity $v(t) = 3t^2$, displacement from $t = 0$ to $t = 2$ is

a. 8 ✓

b. 12

c. 6

d. 4

Applying Integrals principles, $\int_0^2 3t^2 dt = [t^3]_0^2 = 8$. Therefore, the verified correct option is 8.

39 [Challenging] • Target: Visualizing the two parts of FTC together • Cross-Lesson: 2.02 Continuity

Logarithmic integral $Li(x) = \int_2^x \frac{dt}{\ln t} \Rightarrow Li'(x) =$

a. $x/\ln x$

b. $1/\ln x$ ✓

c. $\ln x$

d. $1/(x \ln x)$

Applying Fundamental Theorem principles, FTC Part 1. Therefore, the verified correct option is $1/\ln x$.

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40 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 3.01 The Derivative from First Principles

Use Green's theorem to find the area enclosed by the astroid $x = 1\cos^3 t,\ y = 1\sin^3 t$.

a. π

b. $\frac{3\pi}{4}$

c. $\frac{3\pi}{8}$ ✓

d. $\frac{3\pi}{16}$

$x\,dy - y\,dx = 3\sin^2 t\cos^2 t\,dt$. Then $A = \frac{1}{2}\int_0^{2\pi} 3\sin^2 t\cos^2 t\,dt = \frac{3\pi}{8}$.

41 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Stokes' theorem $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$ generalises FTC to:

a. $\mathbb{R}^3$ surfaces

b. scalar fields

c. **differential forms on manifolds** ✓

d. $\mathbb{R}^2$ only

Stokes/Green/Divergence are all cases of the general Stokes theorem for differential forms.

42 [Challenging] • Target: Net change and the FTC • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For f continuous on $[a, b]$, the function $F(x) = \int_a^x f(t)dt$ is guaranteed to be:

a. differentiable on (a, b), with $F' = f$ ✓

b. only continuous, not differentiable

c. a polynomial

d. constant

FTC Part 1 guarantees F is differentiable wherever f is continuous, with $F'(x) = f(x)$.

43 [Challenging] • Target: Visualizing FTC Part 2 as area • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

By the Mean Value Theorem, find $c \in (0, 4)$ with $f'(c) = \frac{f(4) - f(0)}{4 - (0)}$ for $f(x) = x^2$.

a. 3

b. 1

c. 0

d. **2** ✓

$f'(x) = 2x$ and the average rate is $a + b = 4$; solving $2c = 4$ gives $c = 2$.

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44 [Challenging] • Target: Combining FTC Part 1 and Part 2 in one problem • Cross-Lesson: 6.01 Related Rates

Green's theorem area formula: $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$. Comes from $Q_x - P_y =$

a. 0

b. 1 ✓

c. 2

d. $x + y$

$Q_x - P_y = 1 - (-1) = 2 \Rightarrow A = \frac{1}{2}\iint 1\,dA = A/2 \cdot 2.$

45 [Challenging] • Target: evaluating definite integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

MVT for integrals: $f(x) = x^2$ on $[0, 3]$. Value of c:

a. $\sqrt{3}$ ✓

b. $3/\sqrt{3}$

c. 3/2

d. $\sqrt{3}/3$

$f(c) = \text{avg} = 3 \Rightarrow c^2 = 3 \Rightarrow c = \sqrt{3}.$

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